

Handheld radio deveploment Board
Rev 0.1

 Zettaone Technologies Envision. Thoroughly. Agile. Technological.	
Title: Handheld Radio Deveploment Board	
Sheet Name: Macnica_Handheld Radio Deveploment Board	
Size: C	REV: R0.1
Date Created: Monday, July 27, 2026	Sheet: 1 of 31
Author: Anun P	Reviewed by: <Reviewed By>

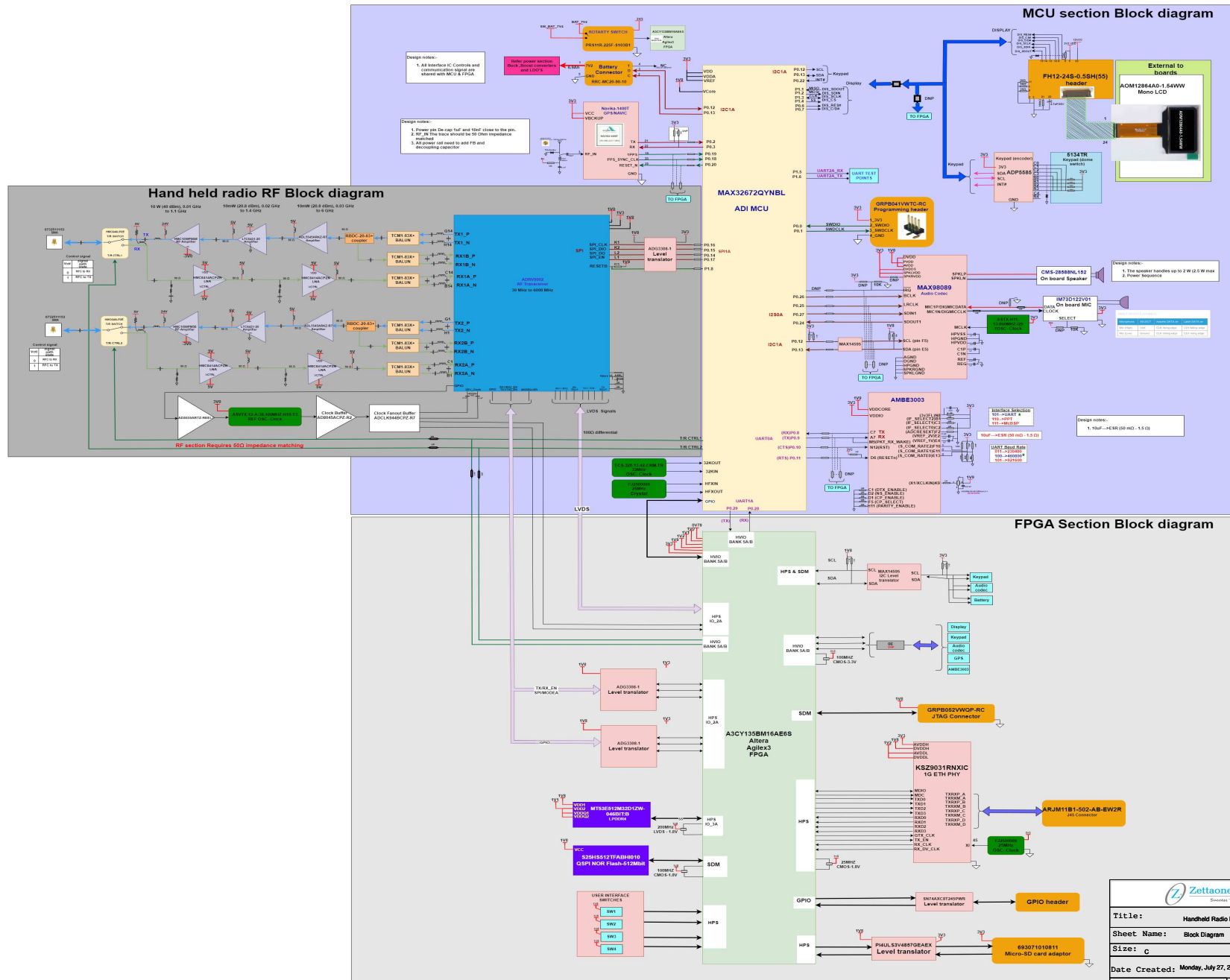
Revision History			
Sl. No	Revision	Description	Date
1	R0.1	Initial Draft	03-07-2027
2	R0.2	Schematic updated based on customer review comments. FPGA Part number is chnaed from "A3CY135BM16AE6S" to "A3CW135BM16A16S" Added 4 test piont for the HPS pins (HPS_IOA_3,HPS_IOA_4, 1.8V(VCC),GND) Connected GTS pins of FPGA to GND through a 0E resistors Connected the user interface switchcpins from "HVIO BANK 5A" to "PLL BANK2A" through the level translator and a 0E resisitor. Connected SPI lines of RF to FPGA by default through a level translator and a 0E resistor Programming header was changed from 5 to 10 pin header(3220-10-0300-00) Spare GPIOs from MCU is connected to "FPGA HVIO bank5A" Changed the resistor(R549) value to 22E Changed the resistor(R550) value to 22E Added 0.1uF capacitor(C831) to the VDD pin of MK1 IC Changed SPKLVD0/SPKRVDD pins of Aduio codec to 5V volt rail Added 22E resistor(R552,R553,R554,R555) to I2S1 lines and made them DNP Changed MCLK input (12.288MHz) to Aduiocodec Added Test points(TP24,TP25,TP26,TP27) to I2S_2 lines IRQ pin of aduiocodec is pulledup 3V3 Changed CP_select pin of AMBE3003 to high(3.3V) Added bulk capacitors (Speaker input LDO) to the 5V LDO output	27-07-2027

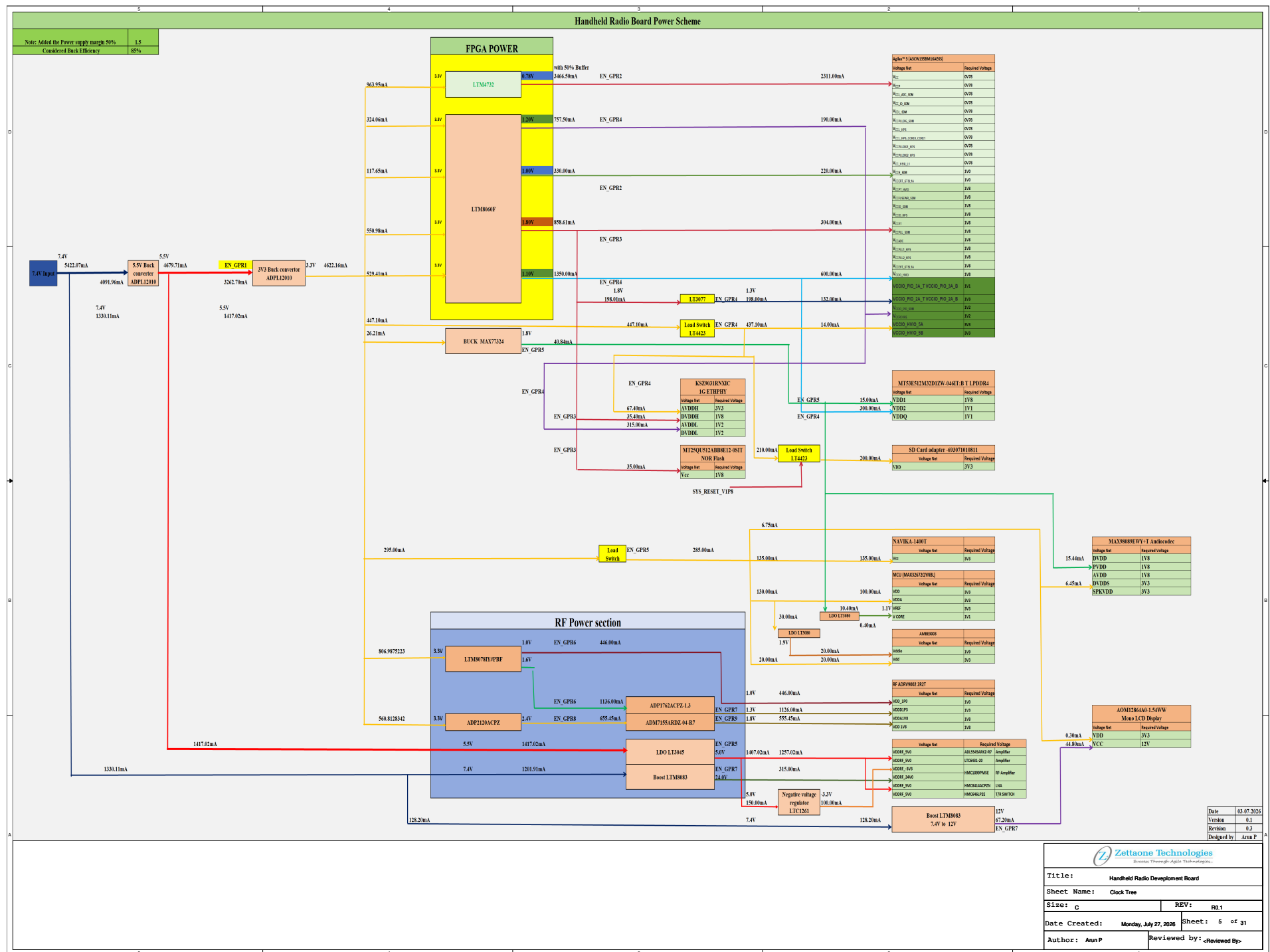
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Hand held radio architecture Block diagram

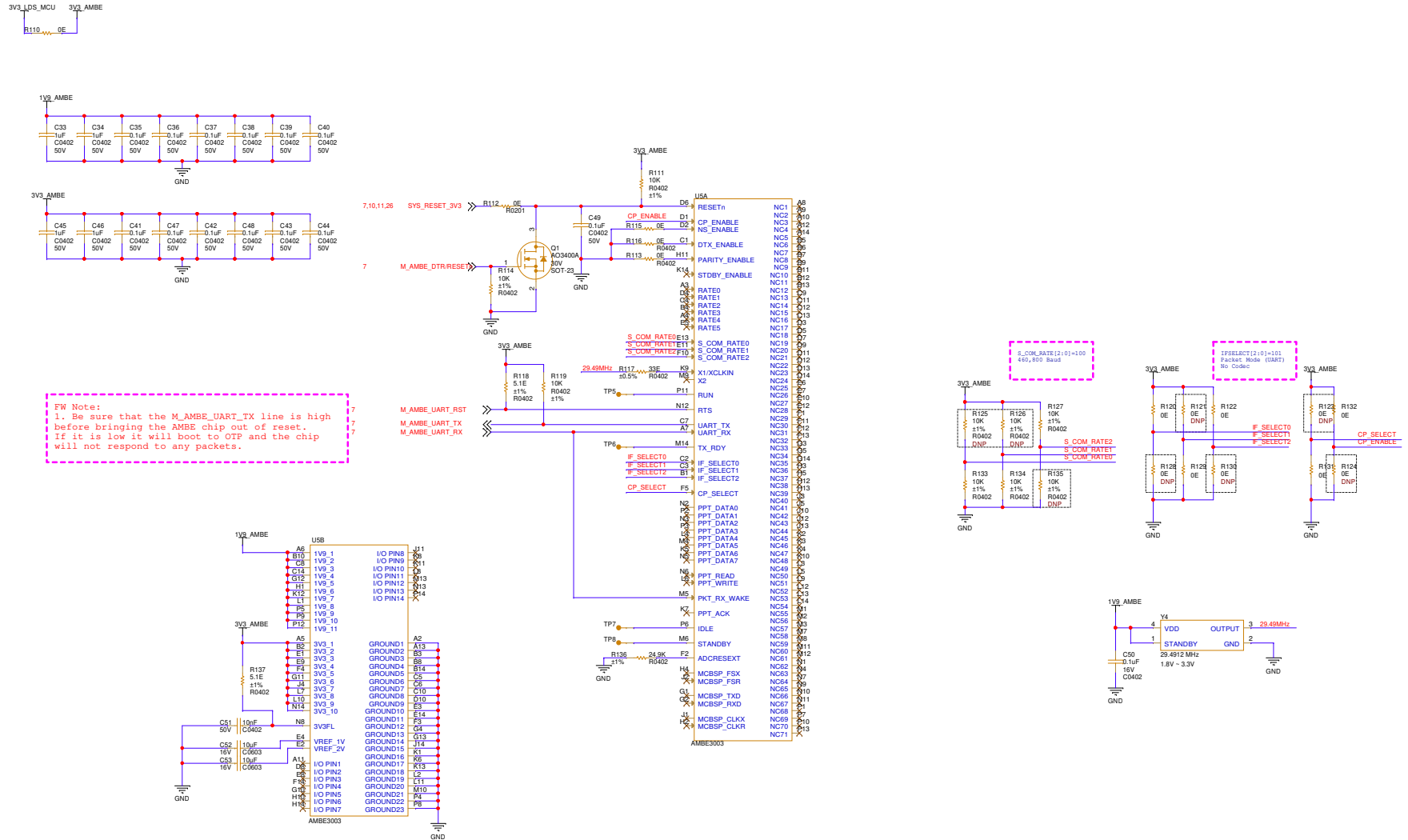




10 W (40 dBm), 0.01 GHz to 1.1 GHz	10mW (20.8 dBm), 0.02 GHz to 1.4 GHz	10mW (20.8 dBm), 0.03 GHz to 6 GHz
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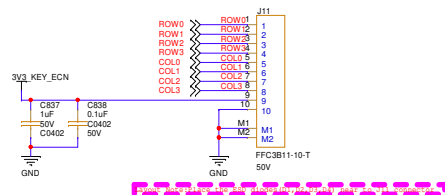


AMBE3003 - VOCODER INTERFACE

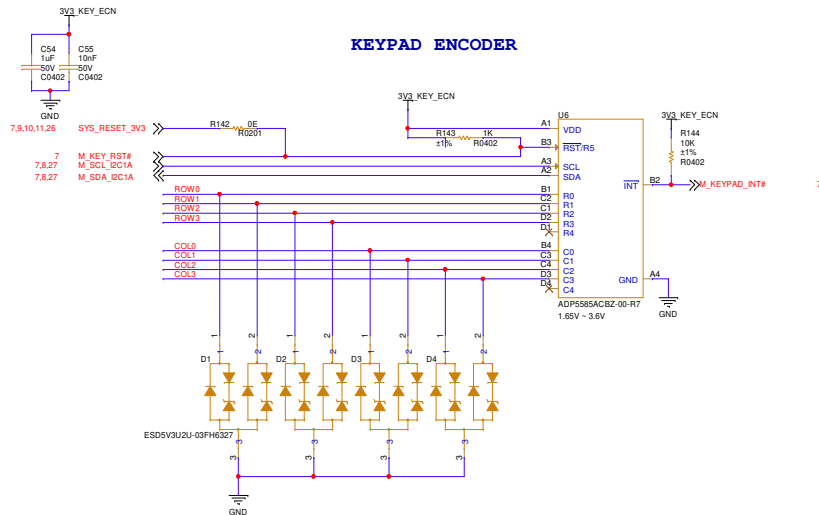


DISPLAY & KEYPAD INTERFACE

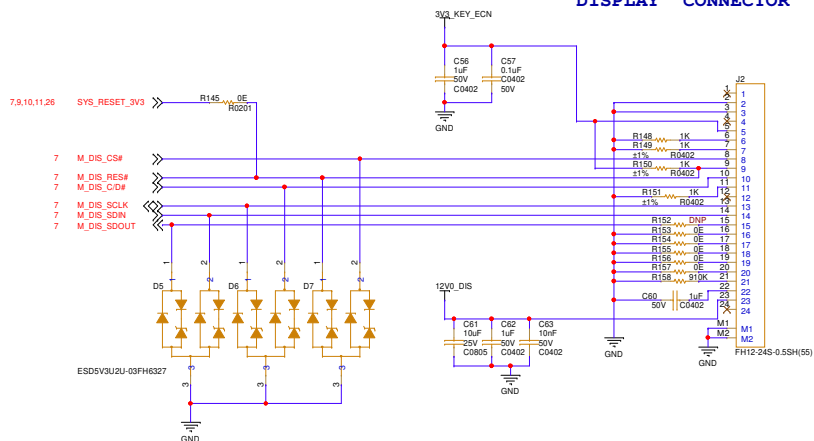
KEYPAD CONNECTOR



KEYPAD ENCODER



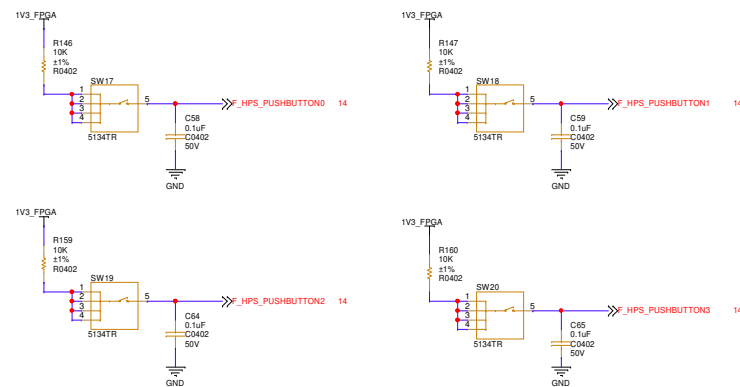
DISPLAY CONNECTOR



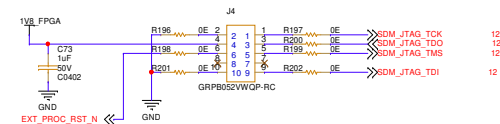
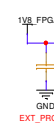
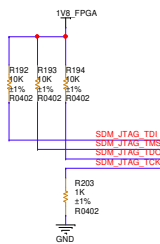
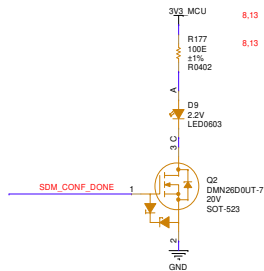
DISPLAY



USER INTERFACE SWITCH

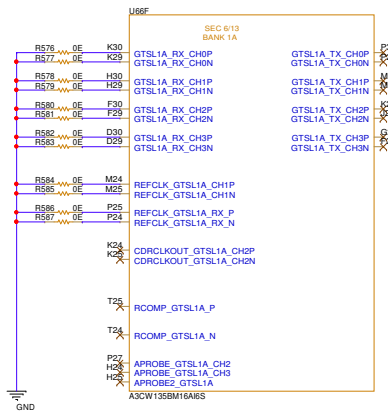
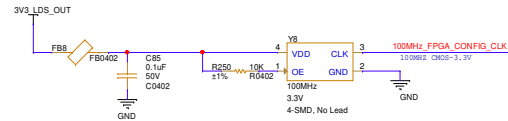
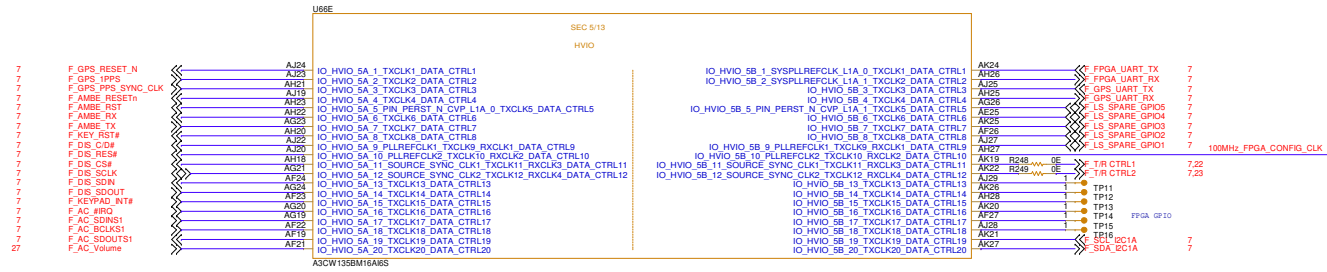


	
Title: Handheld Radio Development Board	
Sheet Name: Display & Keypad Interface	
Size: C	REV: R0.1
Date Created: Monday, July 27, 2020	Sheet: 10 of 31
Author: Anun P	Reviewed by: [Reviewed By]

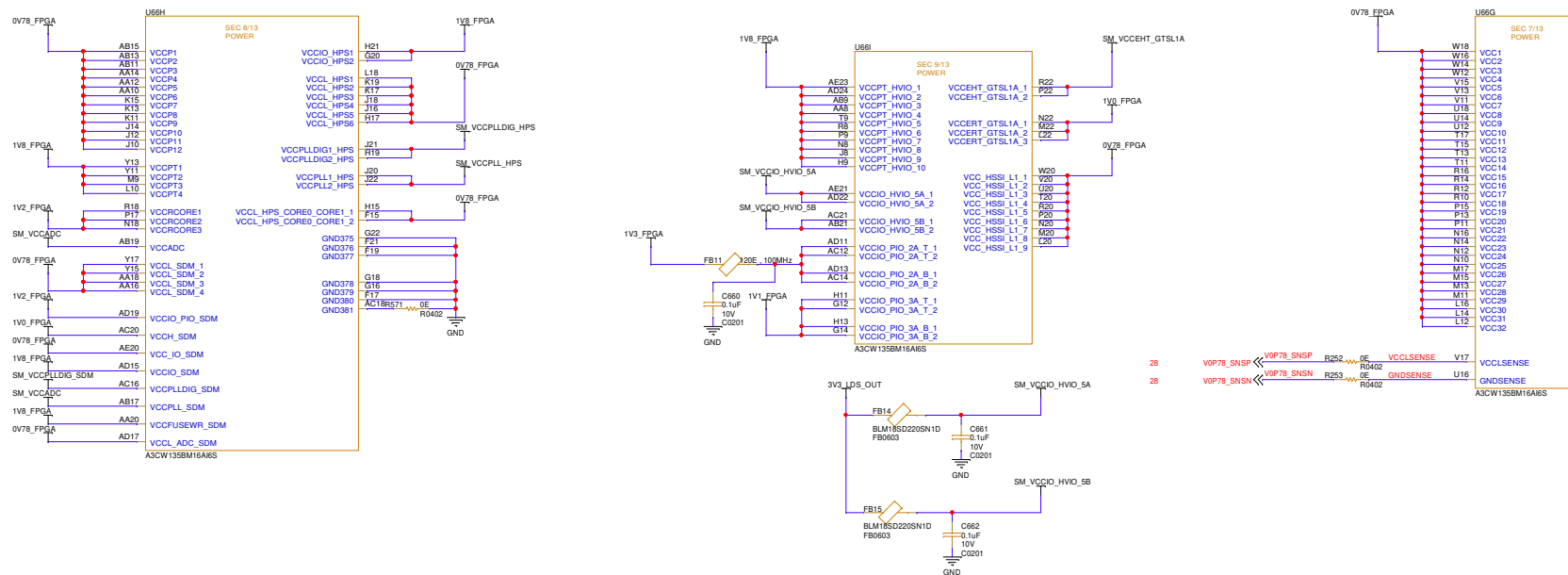


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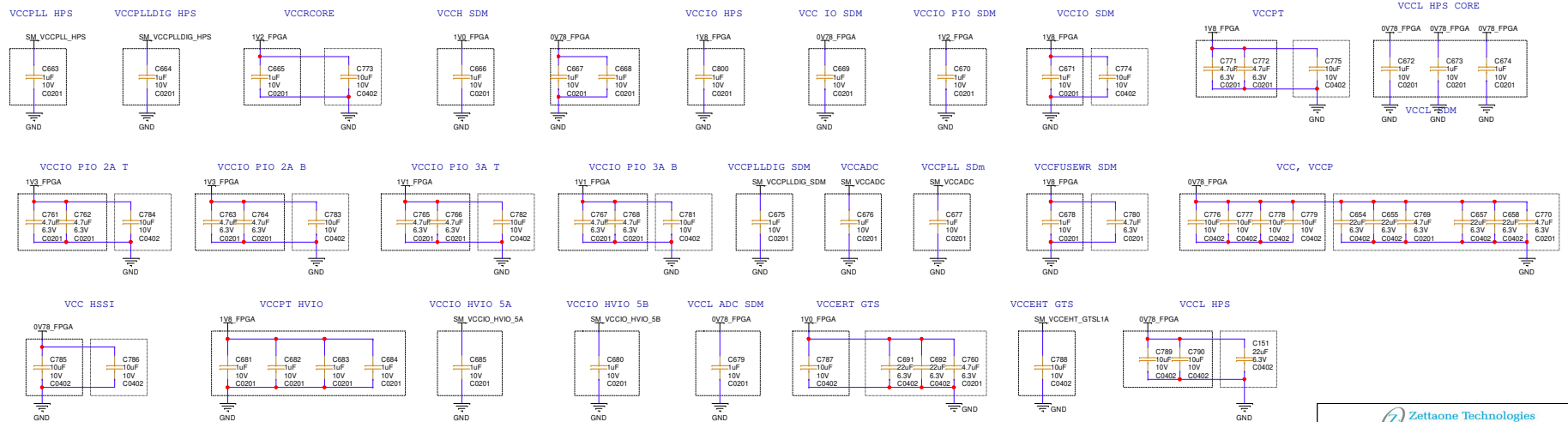
FPGA- INTERFACE



FPGA POWER



FPGA DECAPS

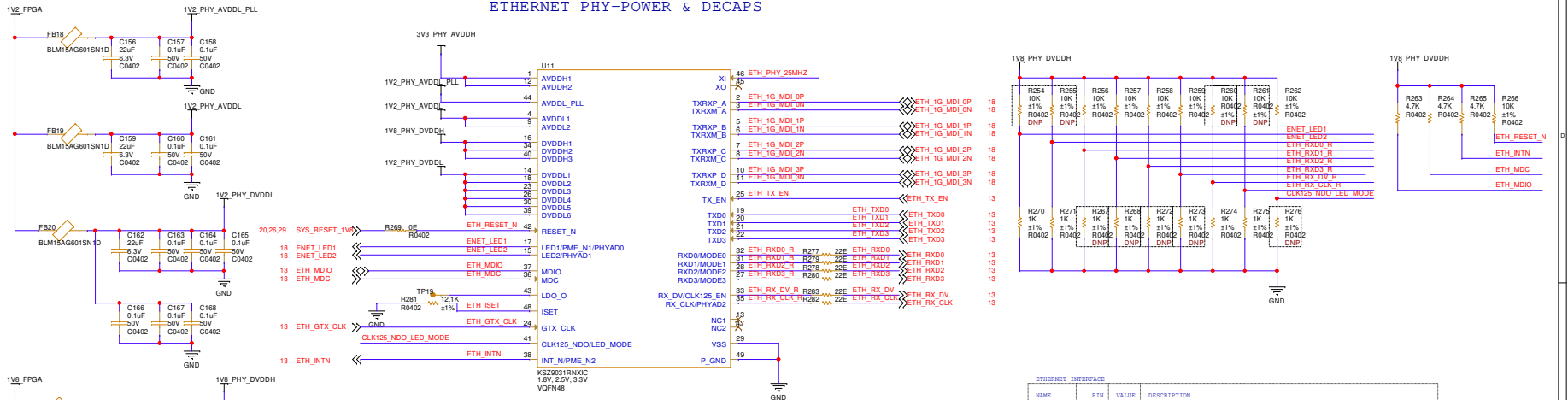


BOTTOM CAPS

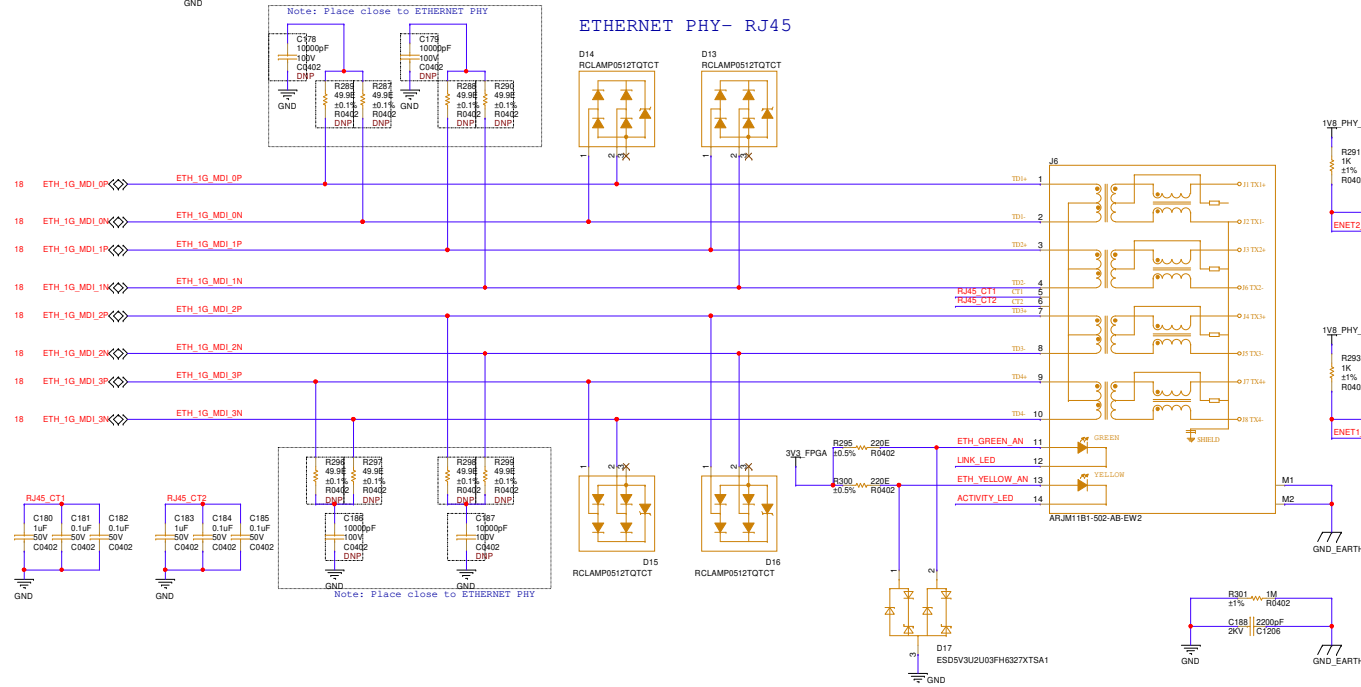
PERIPHERAL CAPS

Title: Handheld Radio Development Board	
Sheet Name: FPGA Power	
Size: C	REV: R0.1
Date Created: Monday, July 27, 2020	Sheet: 16 of 31
Author: Anun P	Reviewed by: Reviewed By

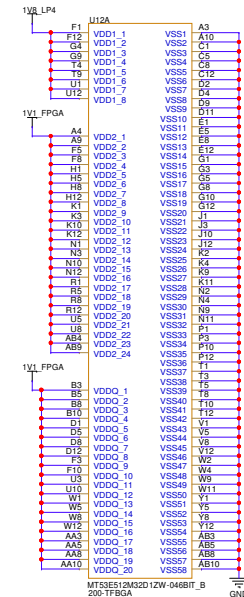
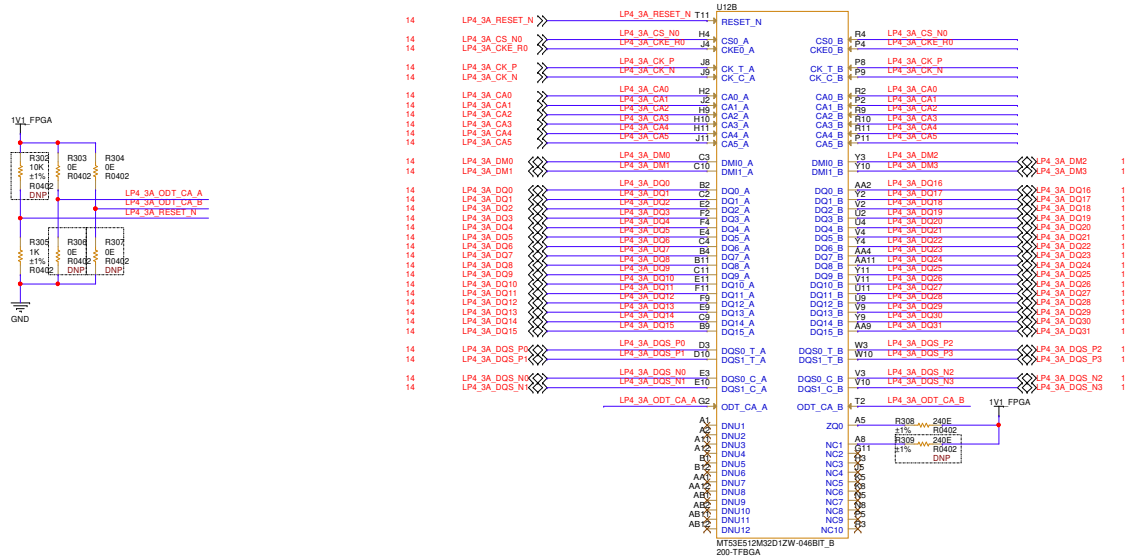
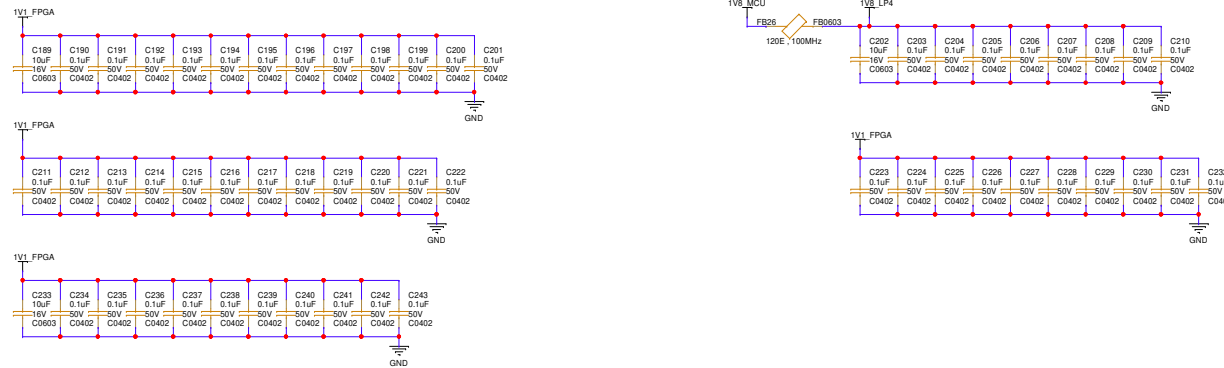
ETHERNET PHY-POWER & DECAPS



ETHERNET PHY- RJ45



HPS LPDDR4 INTERFACE



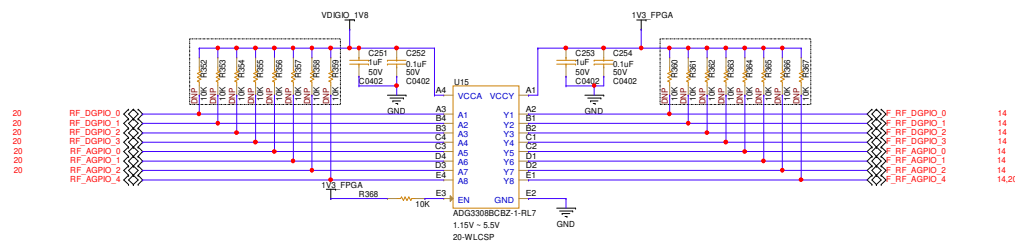
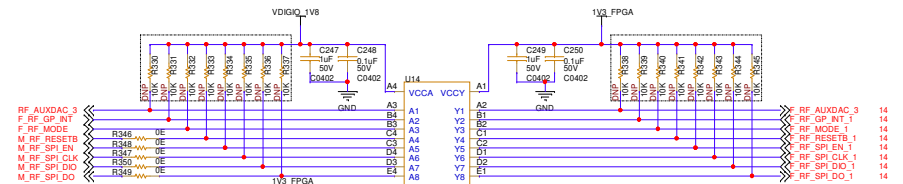
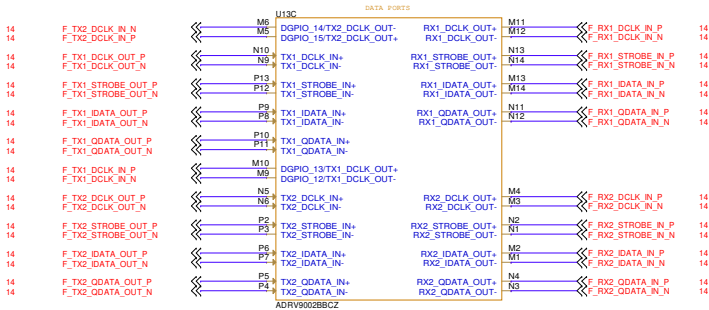
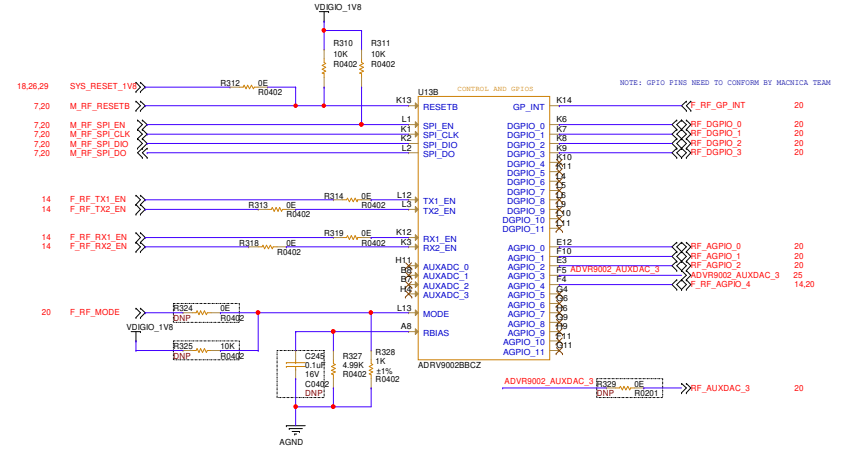
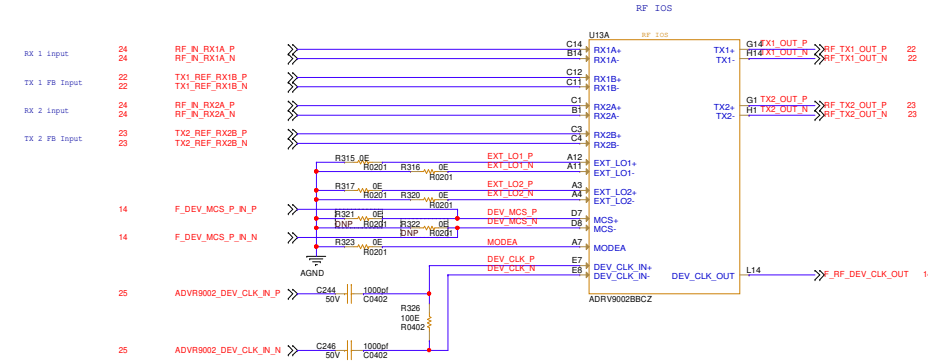
Title:		Handheld Radio Development Board	
Sheet Name:		HPS LPDDR4 Interface	
Size:		C	REV: R0.1
Date Created:		Monday, July 27, 2020	Sheet 19 of 31
Author:		Anun P	Reviewed by: ,Reviewed By,

RF-FPGA INTERFACE

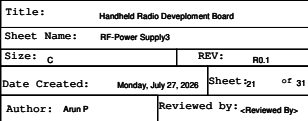
IMPEDANCE CHARACTERISTICS:
ALL DATA PORT DIFFERENTIAL PAIRS = 100ohm DIFFERENTIAL,

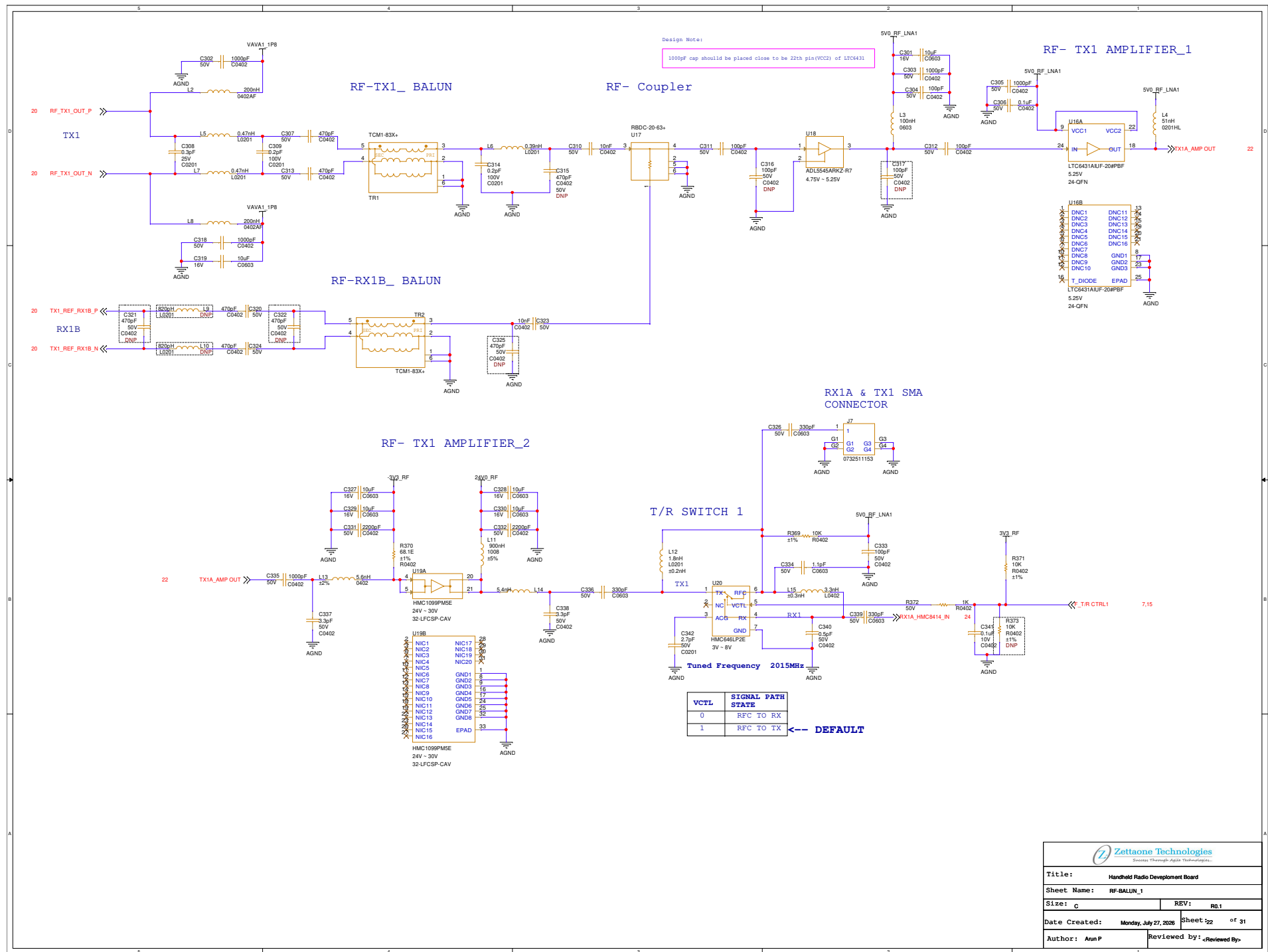
NOTE1: EXT_LO1 AND EXT_LO2 IS NOT USED,
IN THE DESIGN INTERANL LO IS CONSIDERED
NOTE2: IN THE DESIGN MCS IS NOT USED,
THE DESIGN SINGLE ADV9002 IS CONSIDERED.

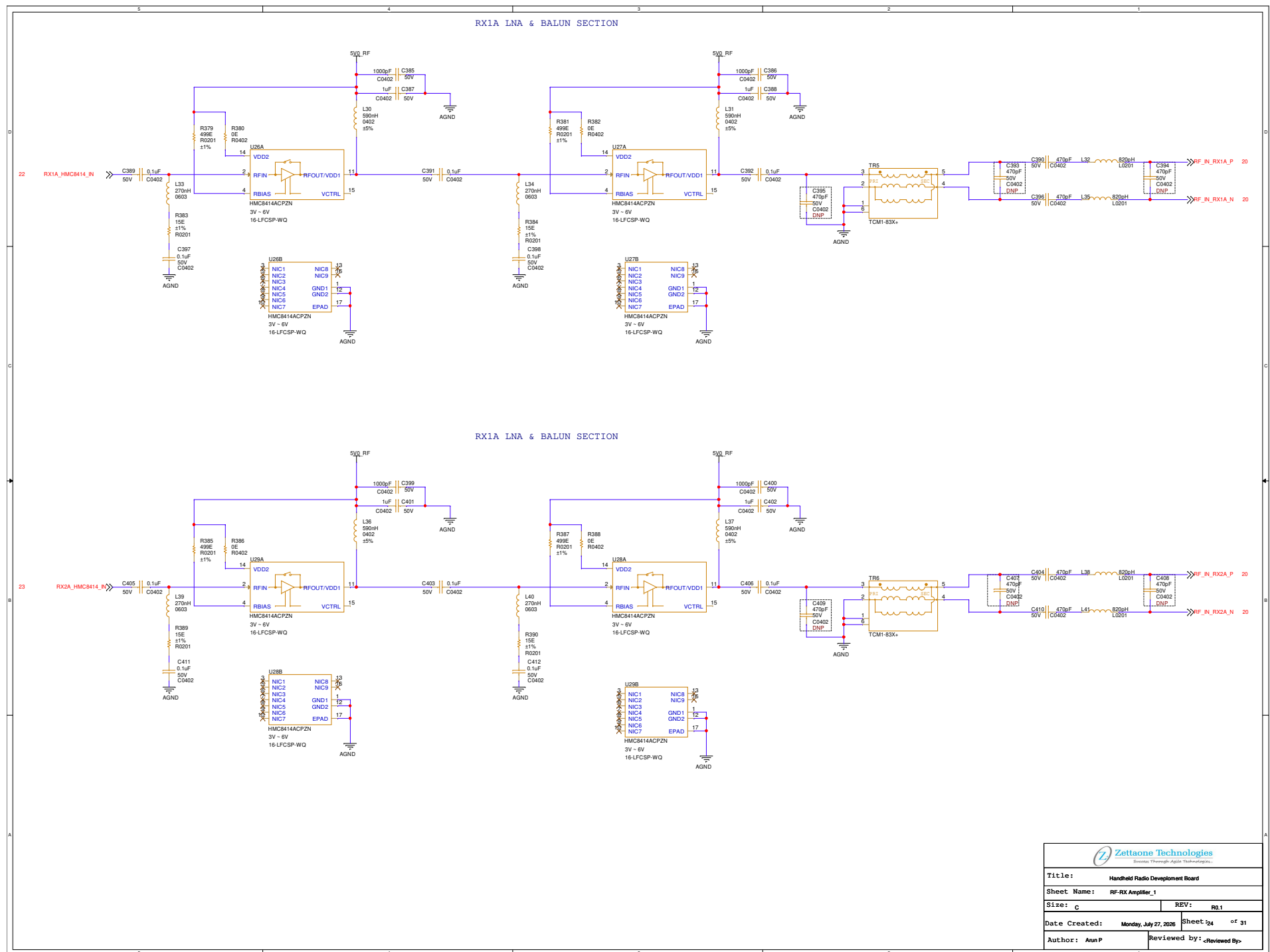
RF-CONTROL AND GPIOs



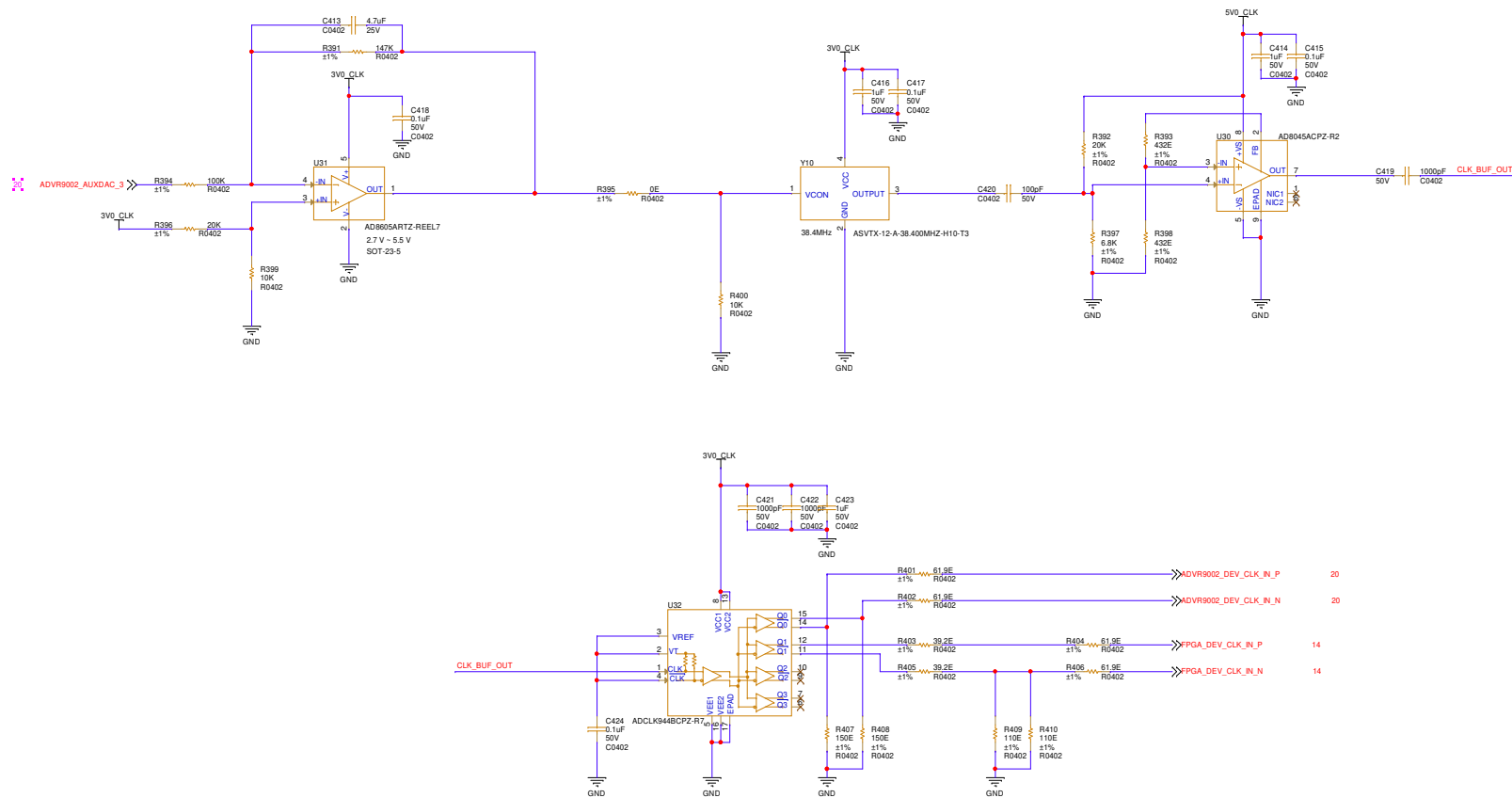
	
Title: Handheld Radio Development Board	
Sheet Name: RF-FPGA Interface	
Size: C	REV: R0.1
Date Created: Monday, July 27, 2020	Sheet 120 of 31
Author: Anup	Reviewed by: Reviewed By

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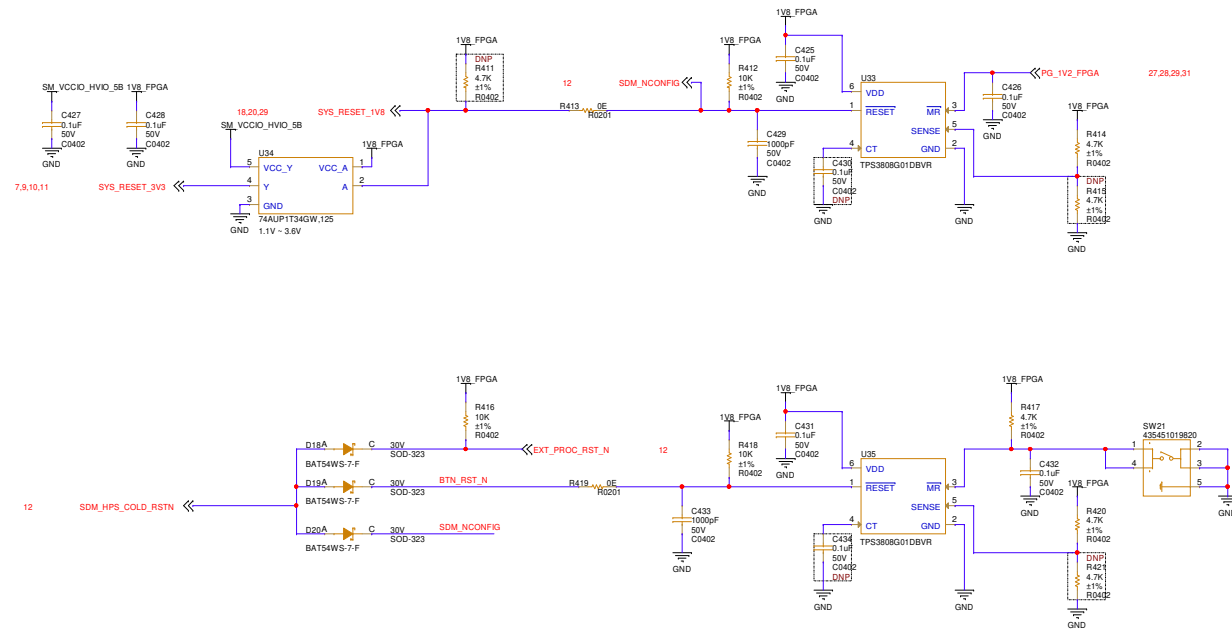


RF-CLOCK DISTRIBUTION

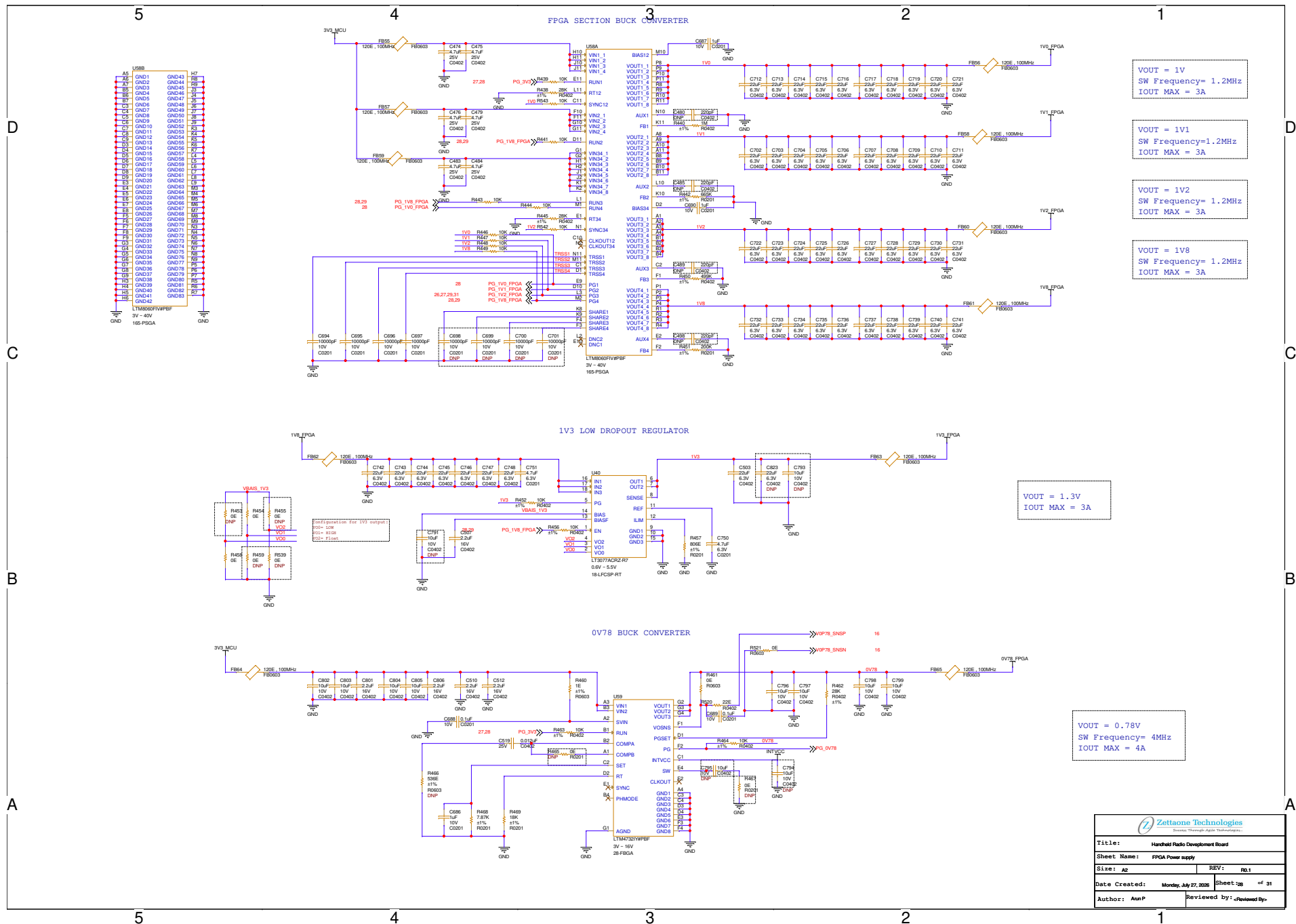


 Zettaone Technologies Solves Through Agile Technologies			
Title: Handheld Radio Development Board			
Sheet Name: RF-Power Supply2			
Size: C		REV: R0.1	
Date Created: Monday, July 27, 2020		Sheet 25 of 31	
Author: Arun P		Reviewed by: <Reviewed By>	

SYSTEM RESET FOR 1V8 AND 3V3

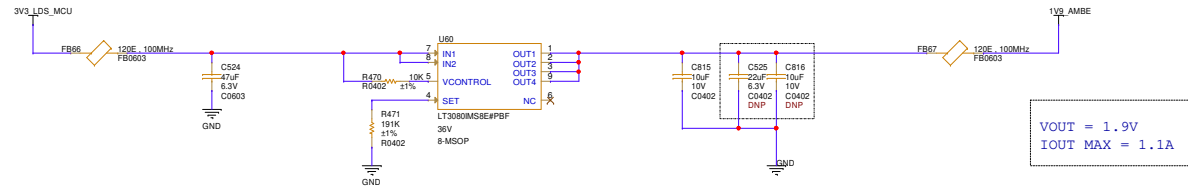


	
Title: Handheld Radio Development Board	
Sheet Name: RF-Power Decaps	
Size: C	REV: R0.1
Date Created: Monday, July 27, 2020	Sheet 128 of 31
Author: Anun P	Reviewed by: <Reviewed By>

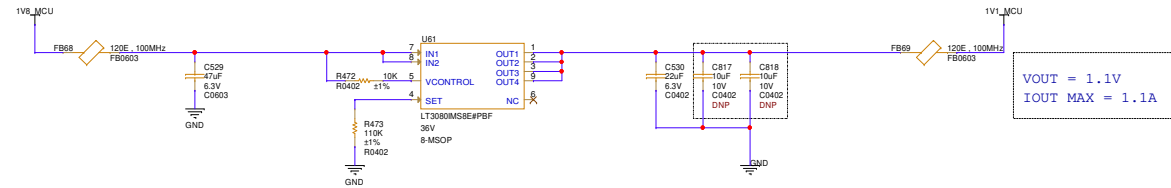


Title: Handheld Radio Development Board			
Sheet Name: FPGA Power supply			
Size: A2		REV: 1.0	REV: 1.0
Date Created: Monday, July 27, 2008		Sheet 28	of 31
Author: AnP		Reviewed by: J. Pineda	

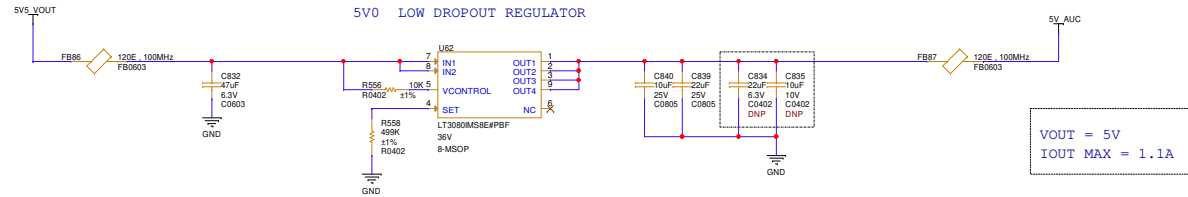
1V9 LOW DROPOUT REGULATOR



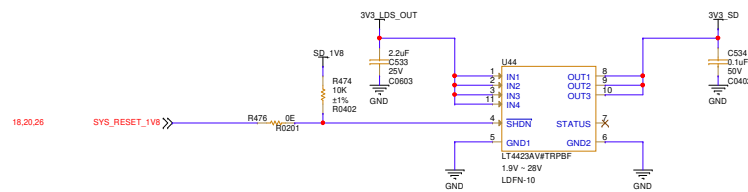
1V1 LOW DROPOUT REGULATOR



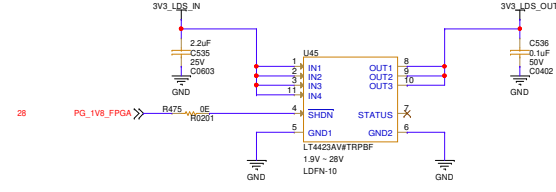
5V0 LOW DROPOUT REGULATOR



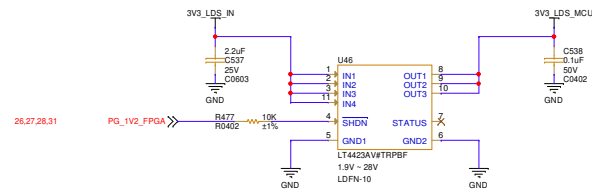
SD CARD LOAD SWITCH



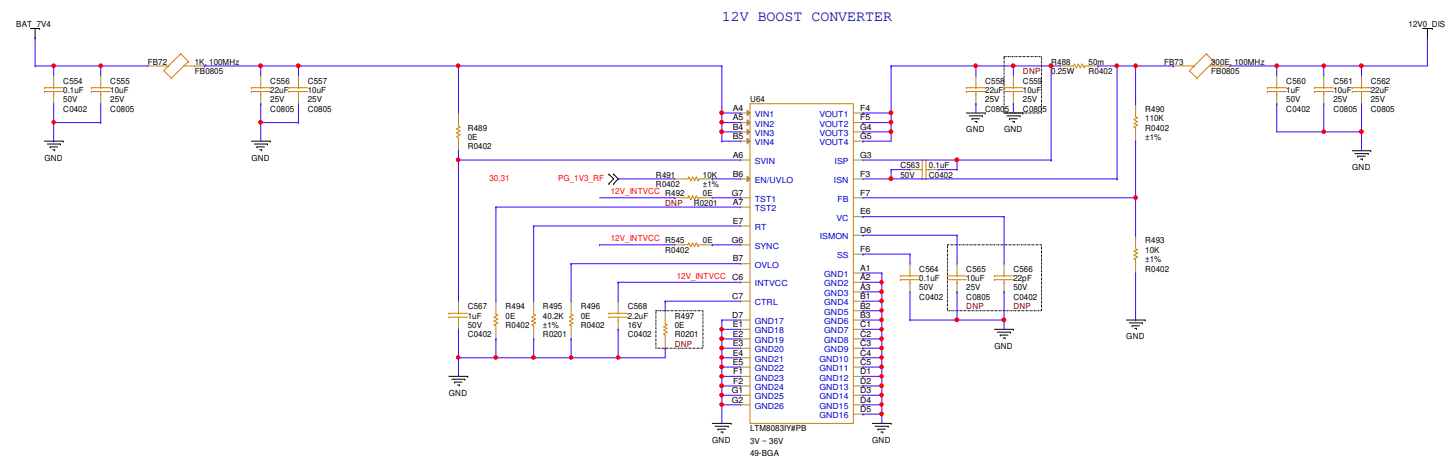
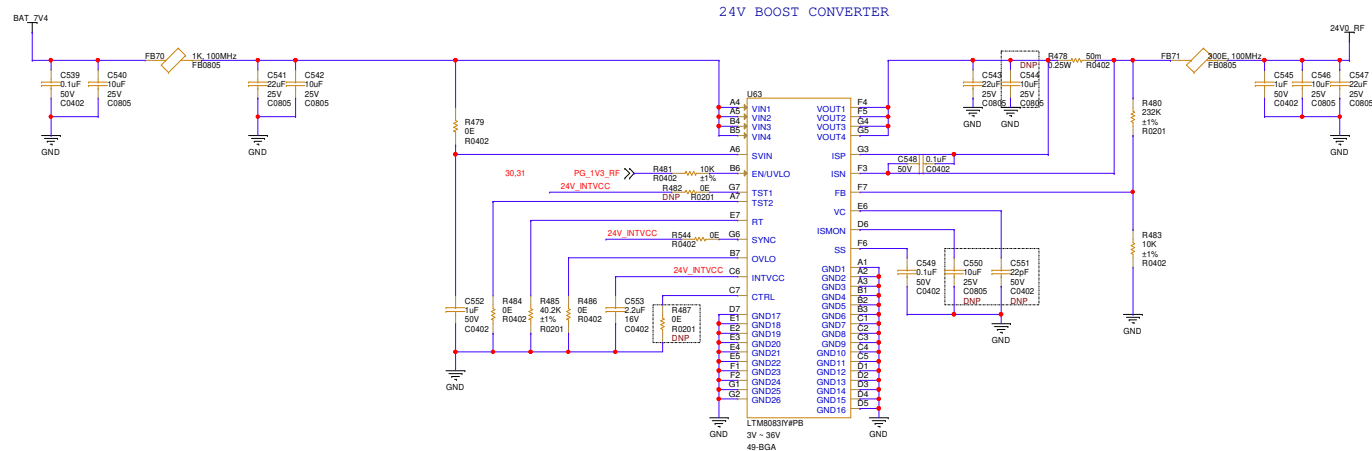
3V3 FPGA 5A/5B BANK LOAD SWITCH



3V3 CONTROLLER LOAD SWITCH



Title: Handheld Radio Development Board	
Sheet Name: MCU & Display Power supply	
Size: C	REV: R0.1
Date Created: Monday, July 27, 2020	Sheet 129 of 31
Author: Anun P	Reviewed by: ,Reviewed By,



Title: Handheld Radio Development Board			
Sheet Name: RF Power supply			
Size: C		REV: R0.1	
Date Created: Monday, July 27, 2020		Sheet 130	of 31
Author: Anun P		Reviewed by: ,Reviewed By,	

